

10 Emerging Technologies To Track In 2010

"We will open the book. Its pages are blank. We are going to put words on them ourselves. The book is called Opportunity and its first chapter is New Year's Day," said Edith Lovejoy Pierce. With similar sentiments for our readers, the electronics industry and us, we are back at our annual exercise of identifying emerging technologies worth watching this year!

It proves to be a very beneficial effort for us as we get a clear picture of what we are likely to be writing about in the next few months. We hope it helps our readers as much—to identify where to concentrate their investments, research, studies and other efforts.

This year, we decided to steer away from the standard buzzwords that you would have heard enough of—3G, wireless technologies, mobile applications, nanotechnology, clean and green technologies, cloud computing, and the likes of those. These, of course, are going to continue in the limelight in the coming year too, but since you are sure to have read quite a lot about these already, let us instead take a look at some newer topics such as printed electronics, through-silicon vias, energy harvesting, etc

■ JANANI GOPALAKRISHNAN VIKRAM

ENERGY HARVESTING

You would have heard this several times already – "We have just one earth; we better take good care of her!" This slogan is becoming louder as the use of information technology and its paraphernalia of gadgets and machines increases. Of the many kinds of environmental damages that our tech toys cause, the power consumed by these devices is a constant worry for the environmentalist and the common man. While the former worries about the resources being depleted to produce this power, the carbon footprint of the devices, the heat released by them, etc, the latter worries about the sky-rocketing energy bills. Adoption of alternative power sources, such as solar power, can be a welcome relief to all concerned parties.

Fortunately, researchers and device manufacturers are making reasonable headways in the field of 'micro energy harvesting'. Macro-level energy harvesting in the form of wind-mills, hydro-electric turbines, etc, have been around for a long time. Now, the trend is towards micro-level harvesting of energy from body heat, vibrations, sunlight, wind, and so on, to scavenge micro-watts of power to run ultra-low power devices.

This positive trend can be attributed to two reasons. One, there has been considerable progress in the field of alternate energy – the constantly

increasing efficiencies of solar cells is a typical example. Two, thanks to superior design and engineering, current generation electronics consume much less power and are amenable to be powered by alternate sources.

Of late, several interesting gadgets are being featured in the media – solar cells to supplement mobile phone batteries, vibration-powered sensors (that are ideal for use in automobiles and other environments where there is a lot of jolting), solar-powered wireless sensor networks, wireless and batteryless switches for use in building automation, heat- or vibration-powered medical implants such as hearing aids and pacemakers that harvest bodily energy using micro-generators manufactured as micro-electro-mechanical-systems (MEMS), and so on.

In fact, an in-body micro-generator that converts energy from the heart-beat into power for implanted medical devices won the Emerging Technology Award at the Institution of Engineering and Technology's (IET) Innovation Awards 2009 held in London, recently.

